



Social network analysis as a tool for the analysis of international trade of wood and non-wood forest products

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ABSTRACT

Countries are almost exclusively seen as the primary unit of analysis in the study of international trade; and analysis of trade in wood and non-wood forest products is no exception to that rule. This paper uses the relationship among countries as the primary unit of analysis and introduces different procedures of social network analysis, where the goal is to demonstrate their applicability to the study of international trade of forest products. Compared to statistical analysis, additional benefit of social network analysis is the fact that the assumption of independence of observations does not hold true; and thus enables a more thorough insight into the structure of the international trade network.

Data on international trade of chemical wood pulp, truffles, and aggregate forest products in the 1988–2006 period is used to demonstrate the applicability of selected network indices as descriptors of economic flows among countries. The analysis covers topics of network visualization, characteristics of the network, grouping of countries, their individual positions and modelling of international trade that includes endogenous effects such as reciprocity, and exogenous effects such as forest certification and contribution of forestry to the gross domestic product.

Results show that many procedures of social network analysis can be used in the field of international trade of forest products, but that great care is needed in their contextual interpretation. Prominent topics of application are the structural trends and impact of different policies, such as the EU Timber Regulation on international trade of wood-based products.

1. Introduction

In the context of globalization, even the critics of global market forces (Hirst and Thompson, 2009) mark international trade as one of the main pillars of world's economic structure. On a micro-economic level, international trade not only provides new business opportunities, it significantly increases employment, total shipments and value-added per worker (Bernard et al., 2007), and within-plant productivity (Bernard et al., 2006). Analysis of international trade has been marked in the last decades with the introduction of methods from different areas of economics (Davis and Weinstein, 2001; Feenstra, 2004), and usage of econometric models (Van Bergeijk and Brakman, 2010; De Benedictis and Taglioni, 2011). However, the biggest change in the

field is the tremendous growth of collection and diffusion of regional or global-level data bases, such as the Comtrade database of the UN's Department of Economic and Social Affairs (2015), Direction of Trade Statistics database of the International Monetary Fund (2006), World Integrated Trade Solutions of the World Bank (2016) and the Eurostat's (2016) database on international trade in goods.

When looking at the statistical procedures used in analysis of international trade (Bhagwati et al., 1998; Feenstra, 2004; Bowen et al., 2012), it can be stated that they are rooted in a substantivist approach, where the basic unit of analysis is the country. As such, they are adequate to describe and draw inferences about parameters that are focused on single countries, such as import quotas and prices, comparative advantage, economies of scale and exchange rates (Krugman,

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2000). This position is contrasted by the relational approach, where the basic unit of analysis is not a country but rather the trade flows between them. For analyzing trade flows structure and the role played by different trade partners, Social network analysis (SNA) has been proposed as a powerful tool. Social network analysis can be defined as “... a comprehensive paradigmatic way of taking social structure seriously by studying directly how patterns of ties allocate resources in a social system” (Wellman, 1988, p. 20). The notion of relation as a fundamental unit of social analysis was later described as the “anti-categorical imperative” by Emirbayer and Goodwin (1994). This puts SNA to close relation with ‘The New York School of Relational Sociology’ and its critique of the dominant ‘substantialist’ approaches to social analysis (Mische, 2011). According to its manifesto (Emirbayer, 1997), SNA is the best way to study social structure from a relational perspective. As such, it is not the preferred way to the study international trade structure from an economists’ perspective as based on country-level parameters. International trade is more than a sum of bilateral trade relations, and SNA can be used to describe and draw inferences about its overall structure and development, as reviewed in the following paragraphs.

SNA have been used to ‘map’ international trade since 1943 (Hilgerdt, 1943), not long after Moreno (1934) introduced sociograms to explain why a pandemic of runaway teenage girls emerged at the New York Training School for Girls in Hudson. Network studies of international trade from the sociology field have focused on position and role analysis of the various market players. The seminal article in this vein is Snyder and Kick’s (1979) description of world systems as a core, semi-periphery and periphery structure, which is based on trade flows, military interventions, diplomatic relations and conjoint treaty membership. Smith and White (1992) followed the same approach and identified five groups following a core–periphery structure based on three time points. Similar studies were done by Kick and Davis (2001) who used two time points and Nemeth and Smith (1985) who used a single time point. Their work was continued by Clark and Beckfield (2009) who applied Borgatti and Everett’s (1999) core–periphery model to describe trichotomous world systems. These procedures were used to assess globalization in international trade by Mahutga (2006) and Kim and Shin (2002). Zhou et al. (2016) focused on the top exporting and importing partners of countries for the 2001–2010 period, and revealed a network focused on the US, China and Germany. They also highlighted a “... need to develop theoretical models explaining, rather than simply describing, the observed networks” (p. 20).

Network studies coming from the fields of economics and physics have concentrated on degree distributions and clustering. Serrano and Boguna (2003) used Comtrade data to map global trade in 2000, and identified features characteristic of large networks: power-law distribution (Barabási and Albert, 1999), small world properties (Milgram, 1967; Watts and Strogatz, 1998) and strong hierarchy (Ravasz and Barabási, 2003). Bhattacharya et al. (2007) assessed symmetrized trade data set against log-normal distribution, and expanded those research directions through dynamical model based on gravity law in Bhattacharya et al. (2008) to show increasing concentration of trade into a few rich countries. Barigozzi et al. (2010) found divergent dynamics between commodity-specific and the aggregate trade network; Barigozzi et al. (2011) assessed the same trade network but focused on the 12 top commodities and found that aggregate networks show globalization trend but the commodity-specific ones do not. De Benedictis and Tajoli (2011) see SNA as complementary to the gravity model, and

compared the two on the same dataset and the same relation was also explored by Fagiolo (2010). Hidalgo et al. (2007) assessed the development of nations through their positioning in ‘product space’, i.e. a proximity matrix of complementary products formally based on their trade patterns, with the assumption that complementary products can be produced and exported with the same type of factors of production, technological sophistication and institutional qualities. This line of investigation was further expanded with the economic complexity index (Hidalgo and Hausmann, 2009), which has proved to be a more accurate predictor of GDP per capita than other frequently used indicators such as global competitiveness index or increased exports of natural resources (Hausmann et al., 2014).

In this paper, we present a series of SNA measures and models coming from the social sciences (see Wasserman and Faust, 1994 for overview), and use them to analyze international trade networks of wood and non-wood based products. First we apply different network visualization techniques as a preliminary analysis of trade structure, then we use descriptive SNA indices to explore the overall network structure and the position of the key countries, and finally we study the network dynamics. The purpose of the paper is not to show an overview of SNA procedures, as they are far too numerous for this type of publication and many do not have relevant interpretation in the context of international trade. Instead, we strive to address the role that prominent SNA metrics can have in understanding the structure and the trends of international trade which cannot be gained from ‘classical’ statistics and models. The rationale behind the selection of each visualization and SNA index is explained following its application.

2. Materials and methods

The analysis was performed on several trade networks, encompassing different commodities and geographical scales. Network visualization techniques were used on global trade network in 2010 of chemical wood pulp (HS code 470200), for which the data is drawn from Forest Products Trade Flow Database (build on Comtrade data; Rougieux et al., 2015) of the European Forest Institute. Descriptive network analysis, that looks at characteristics of individual countries, their grouping and the overall trade network, was performed on the trade of truffles (HS Code 070952) for the 1988–2006 period. Analysis consisted of different SNA procedures that cover the characteristics of the whole network (all world countries and international trade relations), identification of sub-groups and node-level parameters. Chemical wood pulp and truffles are selected as being significative of many opposing forest product characteristics. Wood pulp is a product based on industrial large-scale mass processing technologies, with limited impact on income generation in rural areas but with well consolidated set of traditional key players operating in the international market. Truffles on the other hand are niche luxury product with dynamic international market that is very much connected with rural income generation and small-scale processing. Non-technical explanation of selected network concepts and measures (based on De Nooy et al., 2005 and Snijders et al., 2010) are presented in Appendix A; for detailed explanations, readers are referred to books authored by Wasserman and Faust (1994) and Borgatti et al. (2013).

Network dynamics was analyzed for the cumulative trade of all forest products within EU-27 (i.e. EU 28 without Croatia) for the 1999–2006 period, where the data were drawn from the EFI-WFSE Forest Products Statistics and Trade Flows Data Base (Michie and

Wardle, 2000). Dynamic analysis was performed through application of the Stochastic Actor-Oriented Model (SAOM; Snijders et al., 2010), which was implemented using the SIENA (Simulation Investigation for Empirical Network Analysis; Ripley et al., 2016) computer programme, used through the R software environment (R Development Core Team, 2016). As its name suggest, the modelling focus of the SAOM is on the actors (nodes), who control their outgoing ties, where only one tie change can occur at one point in time (called a micro step), and no coordination in the tie change status is allowed. SAOM also assumes that the current state of the network is conditioned by its previous step, but not the ones before it (i.e. the network is a continuous Markov chain). The model also assumes that individual observations of the network are its states and not brief events, and the change between them is gradual (indicated by the Jaccard index, where values higher than 0.3 for two consecutive observations indicate gradual change). How often an actor can change its network state between two network observations (i.e. number of micro-steps) is defined by the rate function. SAOM also has an objective function, in which actors strive towards a network state in which the value of the objective function is higher – by dissolving a tie, making a new one, or choosing to remain in their current state. This function is set up in terms of probabilities, and is defined by a series of network parameters that influence the probability of changing the network state. SIENA provides parameter estimates and their standard errors. These parameters are called effects, and they can be endogenous (e.g. reciprocity) and exogenous (e.g. country's production capacity, corruption perception index, GDP per capita, existence of bilateral trade agreements, etc.). A positive value of the parameter estimate increases the actor's value of the objective function for the choices that entail the presence of that parameter, and thus increase the probability that an actor will make that change. On example of the reciprocity parameter with positive value, all other things being equal, the focal country will more likely reciprocate a trade relation than engage in trade with another country with which it has not traded before. Convergence of the algorithm is tested by *t*-ratios, were < 0.1 in absolute value indicates good convergence. The analysis was performed in R, using the packages *igraph* (Csardi and Nepusz, 2006) and *rsiena* (Ripley et al., 2016), and in UCINET (Borgatti et al., 2002). Visualizations of the trade networks were prepared using the *igraph* package.

Although all the data used in this paper can be traced to Comtrade, each data set has been treated differently according to its data inconsistency. In the chemical wood pulp data the recorded value of a trade flow is from the reporter which had smaller deviation in unit-price in a five year period, and the missing values are based on regional median unit-price for a given year. Data for the truffle trade were directly drawn from the Comtrade database and 'deep cleaned' before data analysis, according to a procedure developed by the authors, as there is no formally recognized approach for doing it (UN-ESCAP, 2009). Missing data were substituted through the UN Statistic Unit 'mirroring' approach; whenever two countries declare different import-export information for the same trade flow, the record reporting the highest quantity was considered as more reliable; outliers were identified dividing quantities in three categories (small, medium, large), as suggested by Berthou and Emlinger (2011). In the data-set on cumulative trade of all forest products, a single trade flow in a single product was regarded as inconsistent if its unit price was more than five times higher or smaller than one fifth of the average world unit price. Both inconsistent and missing data were substituted based on world average unit prices, and the data set was primary based on trade flows as reported by exporters.

3. Results

3.1. Network visualization

The aim of visualization as the starting point should be to have a quick overview of the whole network, with a focus on the main countries playing key roles at the international level. These insights can steer subsequent analysis, as demonstrated using the example of the trade in truffles. For the example of chemical wood pulp, we adopted four visualization approaches to describe different aspects of the network. The classical way of describing an international market is to state the export and import values of countries that participate in the market. As the basic unit of social network analysis is a tie, Fig. 1 is a graphical presentation of international trade in 2010 on reported exports of chemical wood pulp (HS Code 470200). It represents a classical view of the structure of the market. Export values are shown on the x axis and import values are shown on the y axis. The shade and thickness of the lines reflect the value of the trade flow, and the size of the country's ISO 3 code reflects the size of the exports and imports.

Fig. 1 shows that for China (CHN) and Germany (DEU) trade is dominated by imports, whereas for the USA, Canada (CAN) and South Africa (ZAF) trade is dominated by exports. Although this figure shows which countries are the most prominent in trade of chemical wood pulp, it is very difficult to capture the most valuable bilateral trade flows.

To that end the top 0.3% the most valuable trade flows have been overlaid on a world map (Fig. 2), where value of trade flow is expressed in US\$ million and data are referred to 2010. It can be seen that the most valuable trade flows are Chinese imports from Brazil (BRA; US\$ 412 million), Canada (US\$ 286 million) and Indonesian import from South Africa (US\$ 267 million). These six flows cover 50.8% of the total value of the market, but encompass only 8 out of 87 countries that participate in it; which is far from revealing the structure of global trade.

Fig. 3 uses the same layout but without a map, and displays all trade flows and countries that participate in it, where the size of cir-

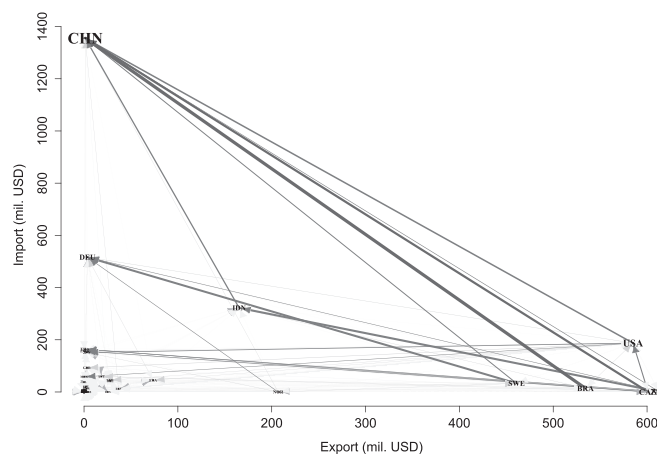


Fig. 1. Global trade of chemical wood pulp in 2010, export/import layout.



Fig. 2. Top 2010 trade flows of chemical wood pulp as overlay of world map (US\$ million).

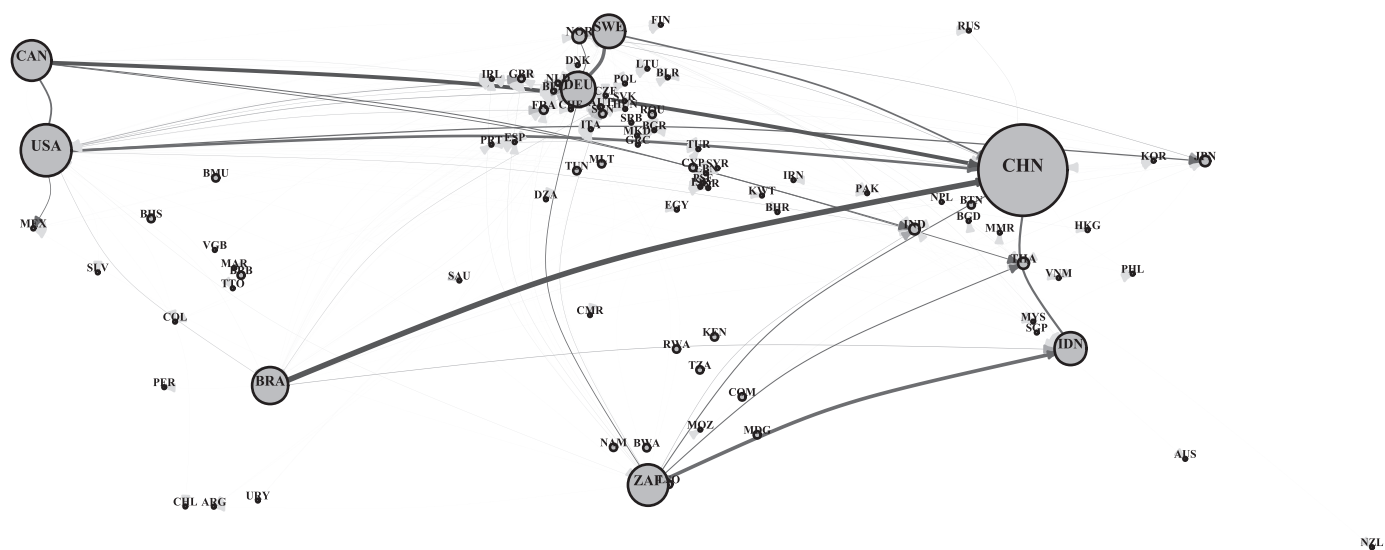


Fig. 3. 2010 global trade of chemical wood pulp in a geographical layout.

cles that represent them and of their ISO 3 codes reflects their total level of exports and imports. The visualization illustrates a dichotomy between the few countries with a major interest in the trade of chemical wood pulp, and the many countries that have a marginal interest in the trade. Fig. 3 also shows that almost all European countries

participated to some degree in the trade, and that Norway (NOR), Japan (JPN) and Thailand (THA) are somewhere between these two groups. Fig. 3 provides more revealing insights into the structure of the global trade as it shows all its elements; however, it is still difficult to see the patterns of trade, and to group countries in some manner.

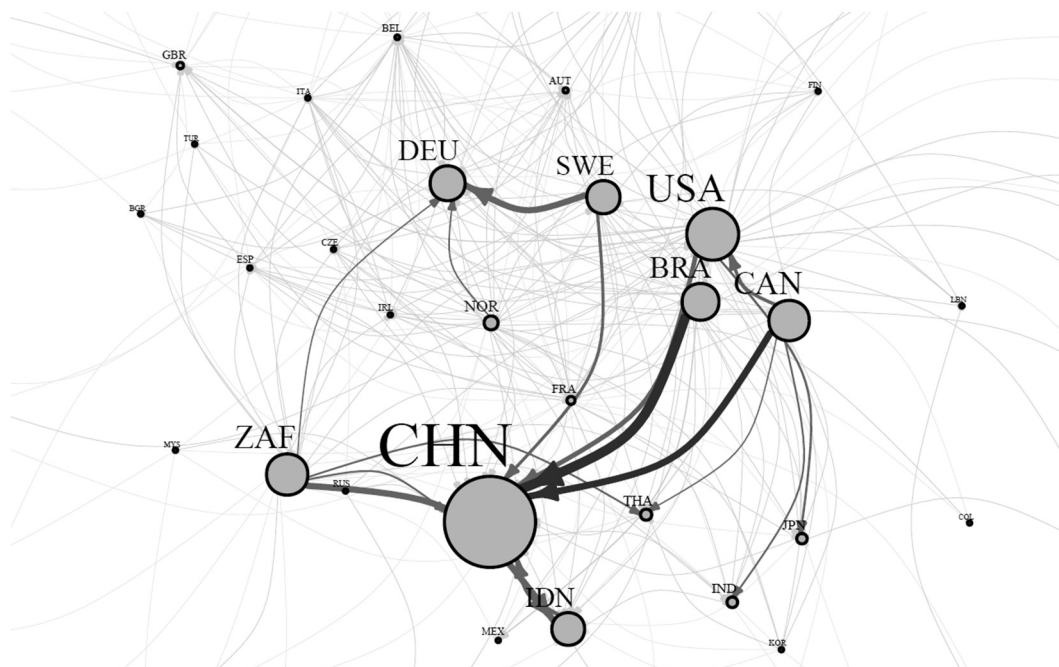


Fig. 4. Central trade relations in 2010 chemical wood pulp trade, Fruchterman-Reingold layout.

Until now the layout of the graph has been based on geographical distances. But how would this map look if it was not based on geographical distances, but on the trade flows? This problem is usually solved with various spring-embedding (or force-directed) algorithms (Eades, 1984; Kamada and Kawai, 1989; Fruchterman and Reingold, 1991; Davidson and Harel, 1996), where ties are treated as mechanical forces that are counterbalanced by repulsive forces with the goal of minimizing distances around the nodes. These repulsive forces can be modified in order to increase or decrease distances in the image, but this does not affect the readability of a graph in which more central, densely interconnected nodes are located in the center of a graph and loosely connected nodes with weak ties are located on the periphery. A Fruchterman-Reingold representation of the 2010 chemical wood pulp network is shown in Fig. 4 (only the central part of the graph is shown; the complete graph including the periphery of the network is shown in Appendix B).

Fig. 4 shows that there is actually no single ‘central’ country, but that each of the ‘core’ countries have their distinct trading ‘neighborhoods’, and that although some patterns can be seen (e.g. a large group of European countries in the top-left section of the graph, and USA, Brazil and Canada being equivalent due to their exports to China) it is difficult to really see the structure of the network beyond several key trade flows. In SNA concepts such as ‘centrality’, ‘core’, (cohesive) ‘groups’, ‘neighborhood’ and ‘equivalence’ have their formal definitions, and networks are primarily studied using matrix algebra which is rooted in graph theory (Harary and Norman, 1953; Harary et al., 1965; Barnes and Harary, 1983). The following sections aim to describe the meaning of these and other SNA concepts in the case of global trade in truffles.

3.2. Descriptive network analysis

An example of general explorative SNA analysis is provided for the international trade of truffles (HS Code 070952) for the 1988–2006 period. The reason behind ending the data series in 2006 lies in the fact that in 2007 there was a reorganization of the HS codes, and truffles were included in the code for “Other mushrooms both wild & cultivated” (070959) HS code. The network for trade in truffles is based on the value of the trade, and contains 202 trading partners.

Fig. 5 shows that the value of the global trade of truffles has grown substantially from the late 1980s until 2006. The global trade in truffles peaked in 1997 at US\$ 61 million, after which the market stabilized between 22 and 31 million. Although the trade network has increased in the 1988–2006 period, it is still very sparse, with binary density of only 0.9% (i.e. the ratio of existing trade relations to all possible ones, or 366 out of 40,602). The diameter of the network (i.e. the maximum length of the trade chain) varied between six and nine over the observed period with no clear trend.

Although almost all of the world countries are involved in the truffle trade, the vast majority of the market is concentrated on few ones. The top 10 trading countries (combined national exports and imports) account for 87% of the total traded value, while the top 20 traders account for 95% of the trade. Four countries (France, Italy, China and Spain) account for 92% of global export value, and five countries (France, Italy, Japan, Germany and USA) account for 70% of global imports (Fig. 6).

By this point, we already have some information on the general structure of the trade network, and we can see that the trade of truffles

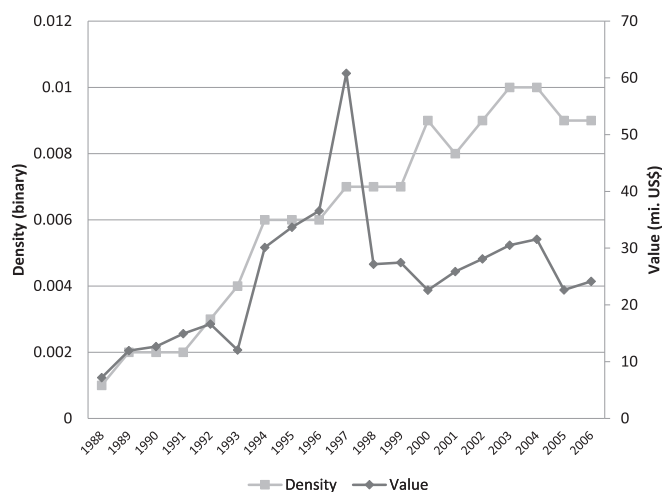


Fig. 5. Value and density of truffle trade, 1988–2006.

has changed a lot over the 1988–2006 period. As we have 19 annual networks, it is somewhat redundant to analyze them all separately. A good starting point is to look for a period of stability within the given time series. This can be done by calculating a ‘proximity matrix’, i.e. a matrix of correlations in Quadratic Assignment Procedure (QAP; Hubert and Schultz, 1976; Krackhardt, 1987) among adjacency matrices (country by country matrix of trade, where rows are senders/exporters, and columns are receivers/importers) of each year (Appendix C). Analysis shows strong changes in the structure of trade in 1988–1989, 1999–2000 and in 2005–2006 periods. Key years which can capture the main changes in the entire time series are 1988, 1995, 2002 and 2006; and they are visualized in Figs. 7–10 using a Davidson-Harel layout (Davidson and Harel, 1996).

At the beginning of the time series, the trade network was very small, and characterized by exports from Italy to Switzerland and the Federal Republic of Germany, and exports from France to Japan. By 1995, the trade network encompasses most countries of the world. Italy is the main exporting country, with France and Germany as its biggest partners. Japan mostly imports from China and not from France, which in turn imports from Italy and Spain and exports to a large number of countries. In 2002, trade became more polarized between Asia and Europe, where France and Germany were the main importing countries, importing mostly from Italy and Spain. France is in very central position

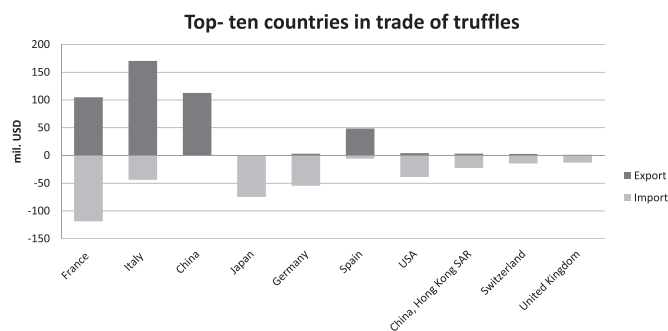


Fig. 6. Export and import of ten ‘top traders’ in truffles, 1988–2006.

in the network with many small exporting ties to other continents. In 2006 exports from France to the USA had greatly increased, and France is acting as a broker between exporters, such as Italy and Spain, and importers from other continents. Imports from China to Japan had strongly decreased, and imports from France to Japan had increased. The main trend in the network, besides the strong overall expansion, is that France has become the key node in trade by establishing cross-continental ties early on.

The expansion of the truffle trade network can be seen by the trends in the change in the number of direct exporting partners (Fig. 11) and direct importing partners (Fig. 12).

Figs. 11 shows that there was a large increase in the number of direct exporting partners for France, Italy and China over the observed period; for China the number of exporting partners decreased from 2002, and Japan and Germany had far smaller out-neighborhood. The number of direct importing partners for the top five countries was much more volatile (Fig. 12); China had a small number of import partners. The main characteristic of the direct trading partnerships of the top five trading countries is that there were many more export partners than import partners.

Looking at the trade networks in 2002 (Fig. 9) and in 2006 (Fig. 10), it is evident that France is in the ‘key’ position. This intuitive notion can be formalized through a key player algorithm (Borgatti, 2006) with the distance-weighted fragmentation criterion, where the ‘key’ node is the one whose removal would ‘disrupt’ the network the most, i.e. cause the greatest lengthening of the average distance between all of pairs of nodes in the network. The greatest fit for the algorithm is when there is only one ‘key player’ – in this example, France.

From a variety of centrality measures (Marsden, 2002), degree centrality and beta-centrality (Bonacich, 1987) are applicable to the analysis of international trade. In our context out-degree centralities equate to levels of exports, and in-degree centralities equate to levels of imports. Beta centrality is a modification of degree centrality, where the contribution of alters (nodes to which the focal node is connected either directly, or indirectly via another node) to the centrality of the ego depends on their own centrality, i.e. are the alters connected to other well-connected nodes or not. The contribution of alters to the centrality of ego diminishes with their distance, and the ‘strength’ of this contribution is set by the value of the beta (β) parameter; where the contribution of alters’ centrality to the centrality of the ego is larger with larger values of the β parameter. If this parameter is positive then the algorithm will attribute higher values to the nodes whose alters are well connected, and if the parameter is negative it will attribute higher values to nodes whose alters are sparsely connected. Both centrality types can be expressed through normalized values, which enables node-level comparison across networks of different size. On example of imports, the difference between degree and beta centrality would tell us if a given country is importing from producers or from countries that have imported this product from some other countries. The association between the top 20 countries and their centrality scores for the truffle trade in 1988–2006 period is illustrated in Fig. 13.

It can be seen from Fig. 13 that France and Italy dominate the market, and that France is both an exporter and importer. There is a small group of predominantly exporting countries, where Spain and Italy mostly export to other countries (i.e. France) who are strong exporters themselves, while China exports strongly to final consumers. Most of the top trading countries are net importers, and they tend to import from other countries which are equally strong and weak importers themselves (i.e. they have very similar I Beta + and I Beta- values). In that group many countries (Belgium, China Hong Kong, Switzerland, USA and Japan) also have small levels of exports.

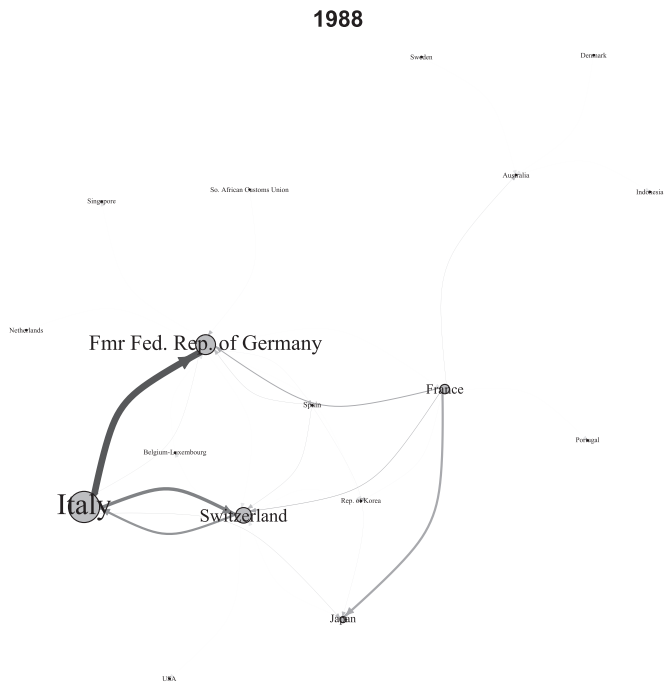


Fig. 7. Global trufflerie trade in 1988.

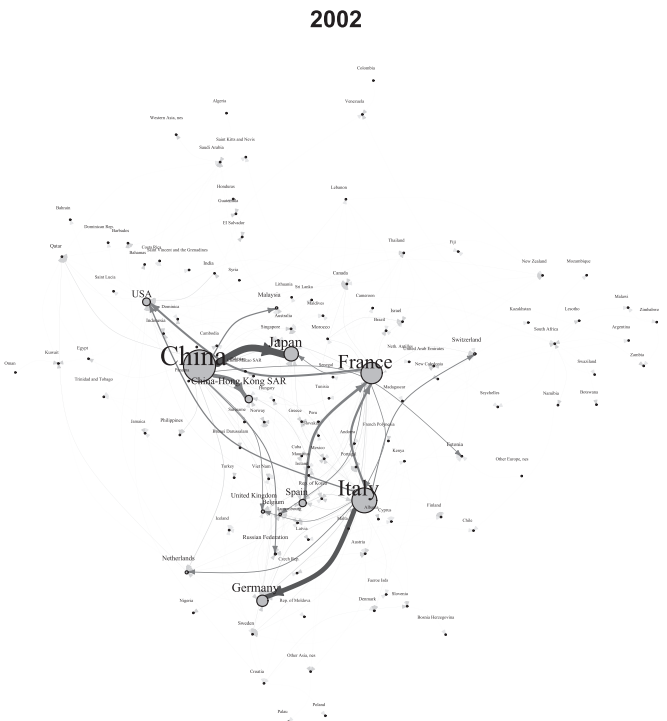


Fig. 9. Global trufflerie trade in 2002.

The figure above shows the ‘centrality’ of countries in the trade network but, although certain clusters of exporters and importers are visible, grouping of countries should be based on analytical procedures that are designed for such purpose. Without dichotomizing and

symmetrizing the data, only two types of analysis have been performed: factions routine (which looks for maximally interconnected groups with minimal connections between the groups) and categorical core-periphery analysis (which looks for maximally interconnected ‘core’ and

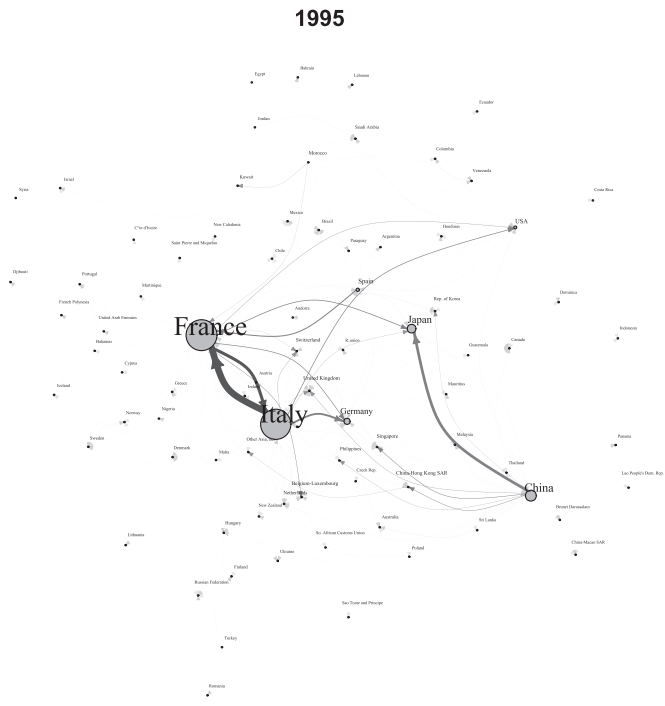


Fig. 8. Global trufflerie trade in 1995.

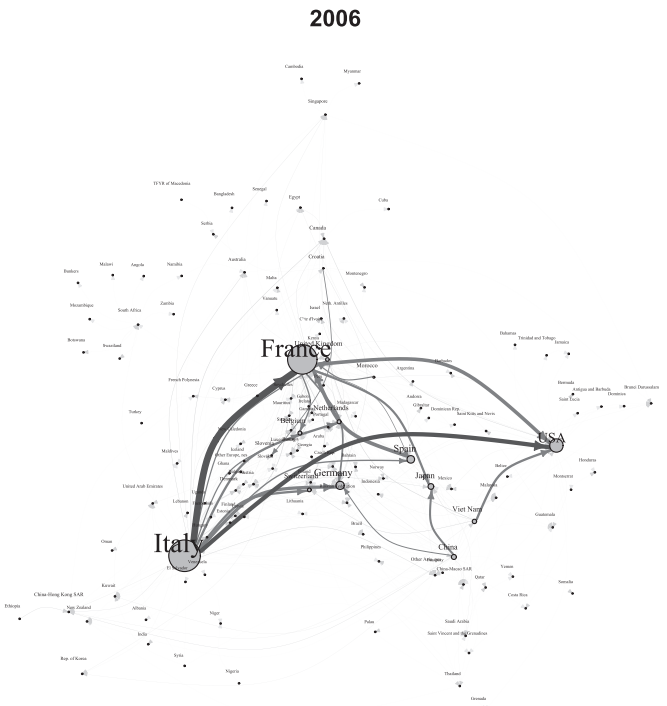


Fig. 10. Global trufflerie trade in 2006.

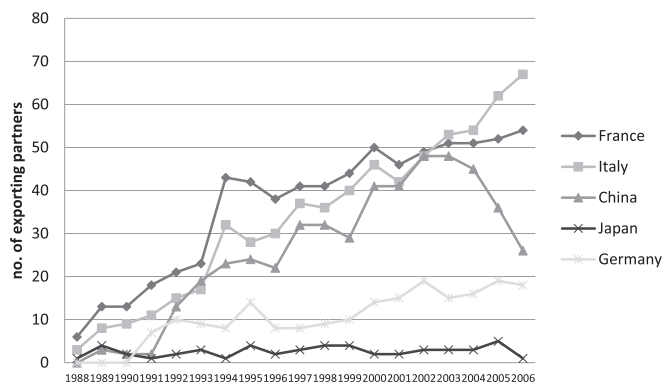


Fig. 11. Number of export partners for the top five countries in truffle trade (1988–2006).

minimally interconnected ‘periphery’). For the overall dataset and 1995–2006 networks Italy and France are in the core. The factions routine for the overall dataset does not provide informative results, as with each added faction single least connected country is removed from the central group – which indicates that there are no distinctive cohesive sub-groups. However, for the 2002–2006 trade networks, the factions routine shows consistent results within minimal badness of fit with three factions. Faction densities for 2006 network are presented in Table 1 and the affiliation of a country to a given faction is shown in Fig. 13.

All European countries except Sweden belong to faction 2, within which most of the truffle trade occurs. In-faction density of US\$ 213,000 actually masks the two dominant trade flows; Italian (US\$ 2.8 million) and Spanish (US\$ 1.5 million) exports to France. First faction depends on French and Italian exports, while from its other trade flows Morocco exports to France (US\$ 474,000), and USA exports to Canada (US\$ 142,000). With the exception of Chinese exports to Germany (US\$ 410,000), faction 3 (Asian countries and Australia) almost exclusively exports within the group, while imports are equally based from faction 2 (i.e. Italy and France) and from within faction 3 (i.e. from China). All Asian countries in the top 20 are in faction 3

except Malaysia and Indonesia. The adjacency matrix of 2006 truffle trade for the top 20 countries with faction affiliation can be seen in Appendix D.

Another way to look at the position of nodes in the trade network is not by density of their ties alone, but by distribution of their ties with respect to a node's attribute (e.g. group affiliation). In a social system marked with exchange or flow of resources, brokerage analysis (Gould and Fernandez, 1989) allows for partitioning of nodes into five distinct roles. On the example of global truffle trade in 2006, the attribute by which partitioning is done is continental position of a country. This analysis can be performed using a weighted or unweighted approach. In the weighted approach, the ego shares the value of a given brokerage relation with all the ‘redundant’ nodes who are in the same type of brokerage role in a tie with the ego's alters. In the unweighted approach, each brokerage role is assigned with full weight. However, reading the ‘raw’ brokerage results may lead to assigning more meaning to it than is justified, as even random assignment of group affiliation may produce brokerage. In order to avoid such interpretations, brokerage analysis produces first expected values (given number of groups and sizes of each group; in our case, the number of continents and number of countries in each continent) and then normalized (or relative) brokerage, which are raw scores divided by expected values given group sizes. Table 2 shows relative brokerage scores for the top 20 countries in valued global trade of truffles in 2006. Given flow 1→2→3, where 2 is the broker, and A, B and C are possible groups, brokerage roles and their explanation is given at the top of its respective column.

As most of the German and Belgian trade occurs within Europe, they have pronounced ‘coordinator’ roles. The Chinese role of Asian ‘gatekeeper’ is visible both from Fig. 10 and the corresponding adjacency matrix (Appendix D), while the same role is played by USA and Canada (where North and Central America are grouped together with Caribbean countries). French exports of European truffles to other continents (‘representative’) have already been commented on, while the same role for Asia is played by Indonesia. Australia acts as ‘liaison’ for its import from France and Italy on one side, and exports to China Hong Kong SAR and Singapore on the other.

By looking at the size and density of French and Italian out-neighborhood (i.e. export) ego networks (Appendix E), it can be seen

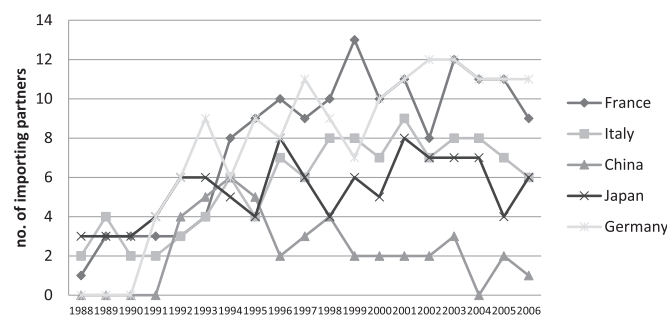


Fig. 12. Number of import partners for the top five countries in truffle trade (1988–2006).

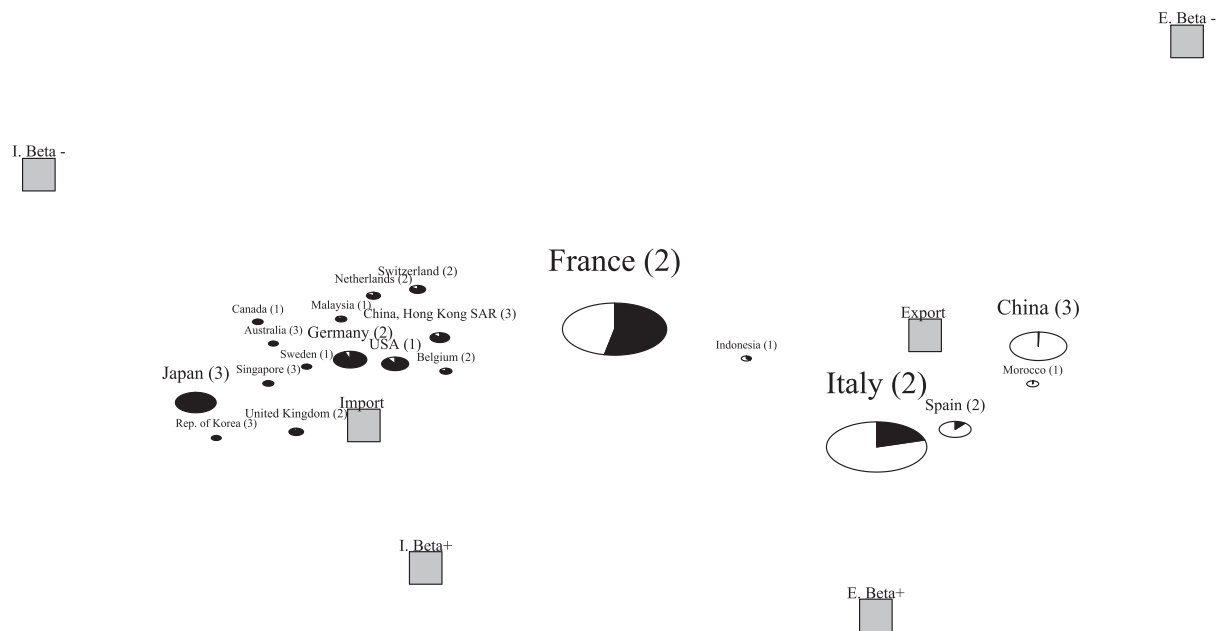


Fig. 13. Association of top trading countries and their centrality scores. Squares represent different centrality measures, and pies represent countries. Export and import correspond to out-degree and in-degree, where + and – beta centrality values are set with largest positive and negative beta values. As beta centrality pays respect to the direction of the tie, standard adjacency matrix of the trade network shows values for export. By transposing the matrices (i.e. switching rows and columns in the adjacency matrix) beta centralities show values for country's import. Following this line of thought, there are four beta centralities; both for Import (I) and Export (E), with maximum (+) and minimum (–) beta value, marked I.Beta+, I.Beta–, E.Beta+ and E.Beta–. The size of the label and the size of the pie reflects a country's total value of trade, and the shaded portion of the pie represents the share of imports, and white portion of the pie represents the share of exports. The number in brackets after a country's name reflects group affiliation as defined by factions routine which is discussed below. The layout of the graph is set by factor values in correspondence analysis (Michael, 2007), which in this context is used to visualize association of two mode data (i.e. countries and their centrality scores). Y coordinates of several truffle-importing countries have been slightly modified in order to avoid label overlap.

Table 1

Density in and between factions for truffle trade in 2006.

No. of faction	1	2	3
Density by faction, US\$ 1000			
1	10.70	11.97	0.01
2	78.53	213.35	31.19
3	1.48	10.58	32.92

Table 2

Brokerage analysis of 2006 truffle trade based continental distribution of countries.

	Coordinator A→A→A	Gatekeeper B→A→A	Representative A→A→B	Consultant B→A→B	Liaison B→A→C
France	4.615	0.700	2.996	0.252	0.253
Italy	4.889	1.080	2.275	0.379	0.304
China	0	4.778	0	0	0.445
Japan	0	0	1.990	1.990	0.742
Germany	11.854	0.971	1.311	0.145	0.036
Spain	9.187	0.707	2.121	0	0.175
USA	1.432	4.468	0.115	0.254	0.267
China, Hong Kong SAR	0	1.706	1.706	0.853	0.636
Switzerland	6.160	1.194	2.389	0	0.222
United Kingdom	3.091	0.963	2.568	0.578	0.359
Netherlands	3.803	0.995	2.765	0.331	0.288
Belgium	13.039	0.758	1.232	0	0.070
Morocco	0	0	0	0	0
Malaysia	8.801	3.413	0	0	0
Singapore	9.583	3.185	0	0	0
Canada	1.866	5.429	0	0	0
Sweden	10.268	0.597	2.389	0	0
Indonesia	0	0	3.484	0	0.927
Rep. of Korea	0	0	0	0	0
Australia	0	0.542	0	0	2.024

Table 3
Analysis of structural holes for 2006 truffle trade.

	Degree	Effective size	Efficiency	Constraint	Hierarchy
France	56	47.104	0.841	0.392	0.631
Italy	68	59.775	0.879	0.336	0.707
China	27	26.026	0.964	0.482	0.606
Japan	7	5.370	0.767	0.582	0.397
Germany	22	14.171	0.644	0.674	0.673
Spain	21	14.138	0.673	0.885	0.725
USA	33	27.226	0.825	0.622	0.695
China, Hong Kong SAR	8	6.876	0.859	0.433	0.264
Switzerland	8	3.638	0.455	1.005	0.598
United Kingdom	20	14.330	0.717	0.657	0.607
Netherlands	18	11.100	0.617	0.935	0.697
Belgium	17	9.987	0.587	0.882	0.630
Morocco	6	3.113	0.519	1.034	0.514
Malaysia	8	7.089	0.886	0.416	0.241
Singapore	9	6.992	0.777	1.146	0.589
Canada	12	9.108	0.759	0.767	0.458
Sweden	10	6.421	0.642	0.813	0.464
Indonesia	8	6.680	0.835	0.740	0.536
Rep. of Korea	3	2.035	0.678	1.132	0.104
Australia	9	5.075	0.564	1.780	0.681

that Italian export network is characterized by larger size and smaller density, especially at the higher levels of dichotomization. Such export network structure of high size and low density may be seen as favorable for the ego country, as it has many trading partners to choose from while they themselves have limited trading options. As mean export price for one kilo of truffles (CN8 code 07095200; Eurostat, 2016) in 2006 for France was €174 and for Italy was €356, this is a

complementary finding. However, these differences are not so pronounced to ‘explain’ the differences in prices on their own. The intuitive notions of favorable and unfavorable trading networks of a single country can be explored through the analysis of structural holes (Burt, 1992 and 2004), which looks at comparative advantages of nodes as a result of their embeddedness in ego networks. Attribute-type outputs of structural holes analysis for valued ties in the 2006

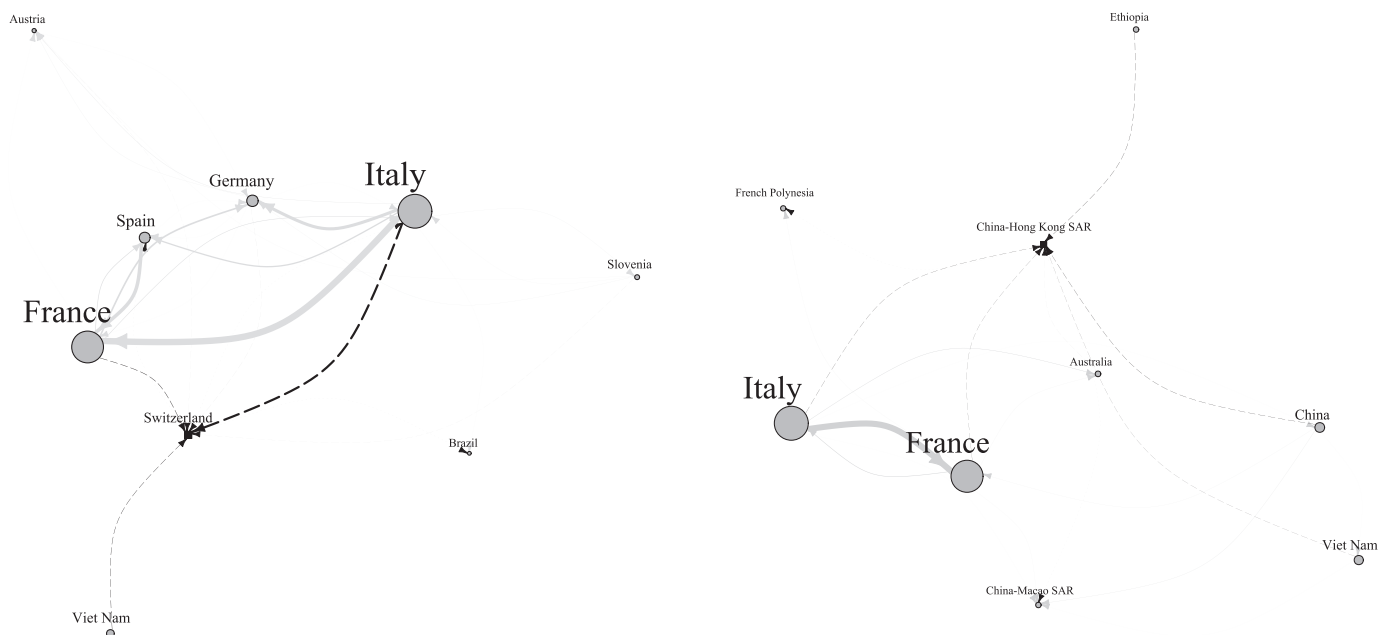


Fig. 14. Ego networks of Switzerland and China Hong-Kong SAR in 2006 truffle trade. Alters are depicted with grey circles; egos (i.e. Switzerland and China Hong-Kong SAR) are marked with a black square; ties between alters are grey; ties between alters and the ego are black. Ego's export ties are marked with dotted lines and ego's import ties are marked with long dashed lines. The size of country symbol reflect its total trade, width of a tie reflects its value, and the graphs are in *Davidson-Harel* layout.

Table 4
Country's attributes used in the SIENA model.

Short name	Description	Data source
CPI	Corruption perception index. Ranges from 0 (highly corrupt) to 10 (highly clean)	Transparency International, 2015
EU_mem	History in EU membership 1(1957–1973), 2(1973–1981), 3(1981–1986), 4(1986–1995), 5(1995–2004) and 6 (2004–)	EC, 2016a.
F_GDP	Share of forestry, wood and paper industry in GDP (%). Values from 2000.	UNECE, 2015
Illegal_T	Import of illegal roundwood, tens of millions of m ³ equivalent volume. Values from 2004	WWF, 2005
VAT	Standard value added tax. Values from 2002	EC, 2016b
FSC	Total number of all valid certificates in the FSC system. Values from 2006	FSC, 2016
PEFC	Total number of all valid certificates in the PEFC system. Values from 2006	PEFC, 2006
Production	Production of all timber based products in 100,000 m ³	FAOSTAT, 2014

truffle trade network are presented in Table 3, whereas these results for different dichotomizations of that network are presented in Appendix F.

Table 3 shows that Italy has both larger size (i.e. degree) and effective size (the number of alters for an ego minus the average number of ties that each alter has to other alters) of its ego network than France. The Italian network is also more efficient (effective size divided by size) than the French network; all of this leads to Italian network exerting lesser constraint on it than it is the case with France, which can be interpreted as Italy having more bargaining power due to the nature of its trade network. The level of hierarchy in the constraint that the ego

network exerts on the ego is somewhat higher for Italy than for France, due to strong Italian exports to France (highest dyadic constraint of Italy is 0.22 by France, followed with 0.04 by USA, whereas for France it is 0.18 by Italy and 0.04 by Spain).

If we compare the position of Malaysia and Indonesia truffle trade in 2006, both of them are in the same faction and have the same size of ego network, but Indonesia is in a less favorable position due to lower effectiveness and efficiency and higher constraint of its ego network. Looking just at the trade network of the top 20 countries, Malaysia is the only country in the faction 1 that exports to members of faction 3, and it also imports from faction 1, which as seen in

Table 5
Parameter estimates of SIENA model.

		Model 1		Model 2		
		13,000 m ³ (median value)		234,000 m ³ (mean value)		
		0.196		0.026		
		Estimate	Standard error	Estimate	Standard error	
Rate function	Rate period 1	1.3951	0.2781***	0.2295	0.1263	
	Rate period 2	1.2344	0.2434***	0.4965	0.2303*	
	Rate period 3	1.2333	0.2447***	0.4187	0.2225	
	Rate period 4	1.3892	0.2609***	0.2518	0.1768	
	Rate period 5	1.2161	0.2564***	0.2386	0.1633	
	Rate period 6	1.4567	0.2847***	0.3604	0.1770*	
	Rate period 7	1.6927	0.3052***	0.6200	0.2824*	
Objective function	Endogenous effects	outdegree (density)	− 1.4740	0.3744***	− 4.6674	0.9856***
		reciprocity	1.2897	0.2116***	1.4438	0.8577
		transitive triplets	0.0511	0.0573	1.2301	0.6087*
		3-cycles	− 0.0897	0.0379*	− 0.3425	0.8540
		balance	0.0795	0.0302**	− 0.6351	0.3108*
		outdegree - activity	0.0345	0.0280	− 0.7299	0.4697
		CPI1 alter	− 0.0111	0.0049*	0.0032	0.0235
	Exogenous effects	same EU_mem	0.6040	0.2336**	1.6395	0.8124*
		F_GDP ego	0.0082	0.0100	0.0179	0.0556
		Illegal_T ego	− 0.0017	0.0161	− 0.0879	0.1097
		FSC alter	0.0007	0.0005	0.0004	0.0015
		FSC ego	0.0004	0.0005	0.0078	0.0040*
		PEFC alter	0.0013	0.0003**	0.0020	0.0008*
		PEFC ego	0.0003	0.0003	0.0023	0.0023
		Production ego	0.0015	0.0006*	0.0094	0.0045*

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

Table 1 is a rare type of trade flow (as most of the trade of faction 1 is imported from faction 2, i.e. France and Italy). This example shows that a country can improve its network position by having divergent import and export partners, even if that network is of limited size. Appropriate test for this proposition would be to look for the interaction of the ego-network effectiveness, efficiency and hierarchy on one side and average import and export prices on the other in an SAOM model.

A good example of the role that structure of the neighborhood network has on the ego is comparison of China Hong Kong SAR and Switzerland. In the 1988–2006 period they were eighth and ninth, respectively, in terms of the value of the truffle trade and had a similar share of import (86% and 82%, respectively). In 2006 they had same size of their ego networks, where Switzerland had imported 10,014 kg and exported 131 kg, while China Hong Kong had imported 3894 kg, and exported 790 kg. Their import prices in 2006 were quite different, as for Switzerland it was US\$ 335.56, and for China Hong Kong it was US\$ 158.95. Their ego networks are presented in Fig. 14.

Both countries import from France and Italy, the dominant countries in global truffle trade. However, Switzerland has very strong import from Italy and has both Spain and Germany in its trade network, which are also major countries in the truffle trade. China Hong-Kong SAR, on the other hand, has its imports more equally distributed between five trading partners, and does not have any other major trading country in its network aside France and Italy. Having trade relations with French Polynesia and Ethiopia strongly decreases the density and constraint of Hong-Kong in its ego network, whereas most of the Swiss truffle trade occurs within Europe among countries which are densely interconnected between themselves. Looking at structural holes analysis on valued truffle network in 2006, it can be seen (Table 3) that the two countries have the same ego network size, but that China Hong Kong has double effective size and efficiency, with half of the level of constraint and hierarchy. A similar pattern can be observed for their in-neighborhoods, as density of Hong Kong's network is half of Swiss network, both on $x > 0$ and $x > \text{US\$ } 4832$ (median) levels of dichotomization (normalized brokerage is 40% higher on both levels of dichotomization; Appendix F). On the highest level of dichotomization ($x > \text{US\$ } 63,476$, i.e. mean tie value), both countries have three importing partners, but Switzerland has a density of 0.33, while there are no ties between China Hong Kong's alters, which causes it to have maximum value of normalized brokerage. These endogenous effects can be related to the difference between Swiss and China Hong Kong's import prices.

3.3. Network dynamics

All of the procedures used so far have focused on describing the network(s) with the implicit assumption that future trade is affected by past trade, but without real statistical significance of the relation between different variables. In order to demonstrate the use of SAOM in international trade of wood and non-wood forest products, an explorative SIENA models were designed. The trade flow is cumulative volume (m^3) of all forest products, and the network consists of all EU 27 member countries. The models encompass the 1999–2006 period, where the Jaccard index ranged from 0.59 to 0.68. Data on export quantity in cubic meters are drawn from the Forest Products Trade Flow Database. The descriptive attributes used in the models are presented in Table 4, and SIENA estimates of international trade

of all forest products within EU27 for the 1999–2006 period are presented in Table 5. In order to see if selected variables have different impacts on the network dynamics depending on the size of the trade flow, two converged models have been designed with different levels of trade flow dichotomization: Model 1 with a dichotomization level of 13,000 m^3 (median trade value in the 1999–2006 period); and Model 2 with a dichotomization level of 234,000 m^3 (mean trade value).

It can be seen in Model 1 that, on average, countries had one and a half opportunity to change their network position per year (the rate functions), and that by the end of the period the rate of change had slightly increased, whereas the rate of change of the higher volume trade in Model 2 is much lower and there are no visible trends. The low rate parameters in Model 2 are an expected finding as countries will more seldom change large trade partners than the smaller ones, and especially as the density (i.e. the proportion of existing ties to all possible ties) is much smaller in Model 2 (2.6%) than in Model 1 (19.6%). Negative parameters of outdegree effects in both models indicate pronounced costs for creation of a tie, i.e. if a trading partner has no specific appealing characteristics or trade relations, the costs of creating a new trade relation outweigh its benefits. In general, the trade relations tend to be reciprocal (Model 1), whereas strong trade relations are not (Model 2). In Model 2 there is an indication (transitive triplets effect) of clustering of trade relations on local (i.e. triadic) level, which in the long run can lead to decreases in the diversity of trade partners. Similar tendencies can be seen in Model 1 through negative 3-cycles effect (tendency towards triadic reciprocity), which is an indicator of a hierarchical trade structure.

The balance effect is positive and significant in Model 1, while it is negative and significant in Model 2. This indicates that in general, countries tend to have the same pattern of trade relations as their trading partners have, but that in case of strong trade relations they tend to trade with partners that have different trade networks. In both models the increase of exports has no predicting value on the dispersion of future exports (outdegree – activity effect). A higher corruption perception index of a country tends to be negatively correlated with imports (CPI alter effect in Model 1), whereas for strong trade relations that is not the case (i.e. in Model 2). Countries also tend to trade more with other countries that have entered EU at the same time (same EU_mem effect), whereas share of forestry in GDP and the import of illegal roundwood have no significant effect on international trade. Having more FSC certificates can increase a country's probability of creating a strong export tie (FSC ego effect in Model 2), whereas having more PEFC certificates can increase a country's probability of creating an import tie, regardless of the volume of the trade flow (PEFC alter effect in Model 1 and 2). Higher values of production of all timber based products affect the creation of export ties in both Model 1 and Model 2, but are much stronger predictor of exports for strong trade relations (as the Production ego parameter estimate is six times higher in Model 2 than in Model 1).

4. Discussion and conclusions

Graph visualization techniques can display different aspects of trade network, where the complexity ranges from focus on individual countries (Fig. 1) to overall structure (Fig. 4). However, no visualization can truly reveal a structure of a network with more than several nodes; and this is almost always the case in analysis of international trade. Complex visualizations also cannot replace basic attribute-

based layouts (such as Fig. 1) or simple charts (Fig. 6), as network analysis of a given data set should only follow and not replace classical statistical analysis. However, they can guide researchers into selecting appropriate SNA procedures (as displayed for the analysis of the truffle trade and Figs. 7–10) that ‘fit’ to a given network structure. It has to be noted that force-directed algorithms are primarily designed for simple undirected graphs, whereas international trade is usually complex and asymmetrical. Such layouts can emphasize different elements of network structure (e.g. presence of certain sub-groups or the centrality of a country) depending on the choice of selected algorithm, and even by changing individual parameters of the algorithm (such as repulsion or node mass). These properties make them useful for highlighting certain results of the analysis, but also make them inappropriate as a standalone basis for making conclusions on the network structure. A formal explanation of these algorithms is presented in Kobourov (2013).

Similar to statistical analysis, use of SNA in analysis of international trade is primarily conditioned by the research question which is posed and by the research design that follows it. This may be problematic for SNA, as it assumes that a major part of the answer lies in the nature of relations between the countries, and not in the characteristics of the countries themselves (i.e. it requires a change from substantivist to relational perspective of the researcher). It also has to be noted that in a broader context SNA also combines substantive and relational variables, and quantitative with qualitative analysis (e.g. Hasanagas, 2004; Aurenhammer, 2013), or can be completely qualitative (see Hollstein, 2011 for overview). The concepts linked to the research question must be operationalized through SNA procedures, which in turn have to have appropriate interpretation in the context of international trade. Some procedures of SNA such as indegree and outdegree have straightforward equivalents in import and export values, but this is an exception. From other measures of centrality, positive and negative beta centralities of trade adjacency matrix and its transposed form have intuitive interpretation of export and import trade relations with more and less ‘central’ trade partners. The same can be stated for left (for import) and right (for export) eigenvalue centralities, which is a very similar measure but much less frequently used in the context of international trade (Vega-Redondo, 2007; Goyal, 2009). Borgatti et al. (2013, p.177) advise the use of beta centralities with high beta values for valued networks instead of eigenvector centrality. From other centrality measures, ‘closeness centrality’ (distance from each node to all others) and ‘betweenness centrality’ (how frequently a node falls into shortest path between all pairs of nodes; Freeman, 1979; Freeman et al., 1991) are more difficult to meaningfully interpret in the context of international trade, although some authors (De Benedictis and Tajoli, 2011; De Benedictis et al., 2013) use it in this context. For grouping of nodes, clique analysis should be based on dichotomized data as there is no generally accepted definition of cliques for valued ties (Borgatti et al., 2013, p.197); and the same can be stated for other ‘bottom-up’ approaches (Hanneman and Riddle, 2005) such as N-cliques and K-plexes. Hierarchical clustering and multidimensional scaling are not appropriate for non-symmetric data (as is it the case with continuous core-periphery role analysis), and most computer implementations of the Girvan-Newman procedure do not allow valued data. From other SNA techniques time-stacked adjacency matrices for a limited set of countries can be subjected to correspondence analysis, by which it would be able to assess change in the level of association between countries through time. The REGE

algorithm (Borgatti and Everett, 1993) is an extension of profile similarity in structural equivalence (‘having same ties to same others’; Lorrain and White, 1971) to regular equivalence (‘having similar ties to similar others’: White and Reitz, 1983; Borgatti and Everett, 1989), and it can be used on datasets with multiple relations. This procedure can be used to classify countries in different roles within the trade network(s), which can then be interpreted through other SNA procedures.

Different SNA procedures exist to test hypotheses related to the impact of endogenous and exogenous effects on the network. Multiple Regression and Logistic Regression Quadratic Assignment Procedure (MR-QAP and LR-QAP; Krackhardt, 1987 and 1988) are regression procedures where the dependent variable is the network, the unit of analysis is the dyad, and independent variables can take a form of a matrix (e.g. past trade network, presence of bilateral trade agreements or geographical distance) or of an attribute (e.g. customs duties and the national production of a given product). MR-QAP and LR-QAP produces coefficient estimates and p values of independent variables as well as R^2 and pseudo- R^2 values analogue to multiple regression models in ‘ordinary’ statistics (in MR-QAP coefficients are interpreted analogue to ordinary least squares regression).

An alternative approach to SAOM in conceptualizing network dynamics is to focus modelling on the ties themselves, and to search for micro-configurations of ties which have a specific contextual meaning for the network under study. This approach is used by p^* models or Exponential Random Graph Models (ERGMs – see: Robins et al., 2007; Robins, 2011; Lusher et al., 2013). A key concept in ERGMs is conditional dependence, by which two ties are conditional dependent if they share a single node. Large parameter estimates can be interpreted as a strong propensity of the network to form a certain micro-structure, given the overall configuration of the model. Snijders et al. (2006) have presented the usage of micro-configuration with more than three nodes, and have widened the conditional dependence to configurations where ties are not connected, but where one tie can affect another ties probability of occurrence. For the use of p^* models in international trade, it is crucial to specify micro-configurations that have a meaningful interpretation. Examples of such configurations in the PNet (Wang et al., 2006) nomenclature are A2P-T (one major importer imports from many intermediaries but not from a major exporter) and AT-C (one major importer imports from many intermediaries and exports to a major exporter), where high parameter estimate of A2P-T could be interpreted as an indication of a hierarchical network structure with a clear distinction between exporters, importers and intermediaries, and high parameter estimate of AT-C could be interpreted as indication of a non-hierarchical trade network. Zhou et al. (2016) state use of p^* models as the next necessary step in the network analysis of international trade.

For the SIENA models that were covered in the results section, models of network dynamics can be expanded to an ordered set of networks with different levels of dichotomization, where it would be possible to identify which effects cause trade relations to emerge, and which effects are related to change from a medium-level tie to strong trade relations. They could also be expanded to include other important exogenous variables, such as tariffs and geographical distance, or can be expanded to co-evolution models (Steglich et al., 2010) of network and ‘behavior’ (i.e. a specific characteristics of actors, in the context of international trade it could be price of a given product, GDP, sawnwood consumption per capita or production capacities). All of these procedures combined can shed light on the structure of international trade,

and within the forestry sector answer some important research question, such as mapping of illegal trade flows, the effects of EU Timber regulation on trade flows, and the impact of Voluntary Partnership Agreements on the trade flows of the partnering countries.

Although appropriate data ‘cleaning’ procedures have been applied in order to minimize the inconsistencies in the underlying data sets, the distribution of these inconsistencies has not been analyzed; and thus results of this paper serve primarily as an illustration of SNA's usage on international trade data. Certain discrepancy of the reports on a single trade flow as provided by importers and exporters can be attributed to costs of insurance and transportation (i.e. difference between CIF - Cost Insurance Freight and FOB - Free On Board). Discrepancies that go beyond the CIF - FOB difference can be interpreted as indicators of an illegal trade flow of wood and wood based products (Brunner et al., 1998; Brack et al., 2002; Contreras-Hermosilla et al., 2003; Chang and Peng, 2015), and aggregating these discrepancies across trade flows and products can be used to ‘map’ illegal international trade. For an exploratory test of EU Timber Regulation (EUTR) on international trade, SNA procedures utilized in this paper could be used to compare the situation before and after individual countries stopped being non-fully compliant to the Regulation. However, it would also be prudent to test for specific changes in the international trade that can be attributed to trade policies such as EUTR and Lacey act. In the context of SAOM models, for the hypothesis of tropical hardwood's substitution with temperate hardwood lumber (Giurca et al., 2013), supportive indications would be negative significant effect of dyadic covariate of tropical hardwood trade in the dynamics of temperate hardwood trade, and for consolidation of trade patterns and partnerships supportive indication would be lowering of the rate function estimate and of the outdegree parameter. Another hypothesis is that the price of a ‘legal’ import from a given country tends to be higher for EU importers than for importers

who don't have appropriate trade measures in place (Prestemon, 2015; Dong et al., 2013). In this case, countries that don't have appropriate trade measures in place and those which have scored low in the ‘map’ of illegal trade could be assigned with an identifying attribute. A simple independent samples *t*-test could be used to compare the distribution of unit-prices for trade flows within this group of countries to the distribution of unit-prices for trade flows between these countries and EU member states. Literature also suggests that EU imports from producing countries will increasingly occur via third parties where there is a weak control of product's origin (Fishman and Obidzinski, 2015), which will then increase the importance of intermediary's legality of imports, as they will have a strong effect on the legality of imports of the final consumers (UCS, 2015). These propositions can be tested by analyzing the network of illegal trade flows (i.e. discrepancies in reporting on trade), where supporting findings would be decrease in indegree centrality and increase in positive beta centrality of transposed trade matrix, as well as decrease of left eigenvector centrality of EU countries. Other supporting indications to the hypothesis could be strong lowering of the 3-cycle effect in the SAOM model of EU countries' trade network after EUTR is introduced, and strong increase in the representative role of the brokerage analysis for several countries that have scored low in the ‘map’ of illegal trade for flows that end with EU member states.

Acknowledgements

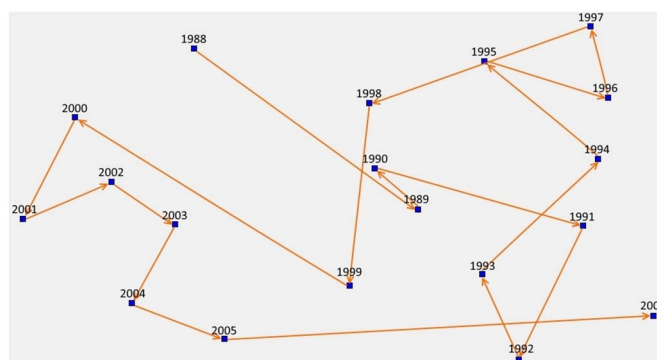
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Appendix A. Non-technical explanation of some network concepts (based on Snijders et al., 2010; De Nooy et al., 2005).

3-cycle	Effect in stochastic actor-oriented models; $A \rightarrow B \rightarrow C \rightarrow A$. Tendency towards triadic reciprocity. Negative effect is in an indicator of local hierarchy
Alters	Nodes to which the focal node (ego) is connected, i.e. the trading partners of the ego. Can also mean indirect trade partners, but then the distance between the ego and the alter has to be stated
Balance	Effect in stochastic actor-oriented models; defined analogue to Burt's concept of structural equivalence with respect to out-ties
Beta (β) centrality	Measure of the centrality of a node. If β is positive, assigns positive weight on the centrality of alters; i.e. calculates how “central” the ego is with respect to ‘centrality’ of its alters. If β is negative, assigns higher centrality to ego whose alters are sparsely connected, and lower centrality to ego whose alters who have many ties. See Bonacich (1987)
Brokerage	Brokerage analysis measures the position of an ego in a network with respect to a given categorical attribute of nodes. Given flow $1 \rightarrow 2 \rightarrow 3$, where 2 is the broker, and A, B and C are possible groups, the options are: Coordinator: $A \rightarrow A \rightarrow A$ (all nodes belong to same group) Gatekeeper: $B \rightarrow A \rightarrow A$ (source belongs to different group) Representative: $A \rightarrow A \rightarrow B$ (recipient belongs to different group) Consultant: $B \rightarrow A \rightarrow B$ (broker belongs to different group) Liaison: $B \rightarrow A \rightarrow C$ (all nodes belong to different groups) See Gould and Fernandez (1989)

Constraint	A summary measure that taps the extent to which an ego's connections are to others who are connected to one another. If an ego's potential trading partners all have one another as potential trading partners, that ego is highly constrained. If an ego's partners do not have other alternatives in the neighborhood, that ego is not highly constrained. See Burt (1992)
Core--periphery	A model of systems that groups the nodes into a core which is “maximally” interconnected, and a periphery, where the nodes are more “loosely” connected See Borgatti and Everett (1999)
Clique	A subgroup in which all possible ties are present
Ego	Focal node; i.e. a country being discussed
Density	For dichotomous data it is a portion of all possible ties. For a valued network, density is defined as the sum of the ties divided by the number of possible ties (i.e. the ratio of all tie strength that is actually present to the number of possible ties). Normalized values can be compared across networks of different sizes
Dyadic constraint	The Dyadic constraint is a measure of the extent to which the relationship between an ego and each of the alters in ego's neighborhood “constrains” ego. A full description is given in Burt (1992) . At the core, A is constrained by its relationship with B to the extent that A does not have many alternatives (has few other ties except the tie to B), and A's other alternatives are also tied to B. If A has few alternatives to exchanging with B, and if those alternative exchange partners are also tied to B, then B is likely to constrain A's behavior
Effect	Explanatory variables within the SIENA model for the network change
Effective size of the network	The number of alters that ego has minus the average number of ties that each alter has to other alters
Efficiency	The effective size of an ego's network divided by its actual size. That is, what proportion of that ego's ties to its neighborhood are ‘non-redundant’
Faction	A type of cohesive subgroup. Ideally, all ties for a faction are within the group and there are no ties outside the group
Hierarchical clustering	Routine for defining cohesive subgroups in which they are defined by an ever increasing level of dichotomization
Hierarchy	Measure that describes the nature of the constraint on an ego. If the total constraint on an ego is concentrated in a single other actor, the hierarchy measure will have a higher value. See Burt (1992)
Indegree	Measure of ingoing ties (number of ties or cumulative strength)
Jaccard index	Measure of similarity between sample sets (i.e. intersection/union). Has values between 0 and 1 (1 marks absolute similarity of sets)
Neighborhood	Alters of an ego in a given distance of ties, where first step neighborhood is marked with + 1, two-step neighborhood is marked with + 2, etc. Directed ties can be split into in-neighborhood and out-neighborhood, which marks the direction of the ties of the focal node. In this context, neighborhood refers to direct trading partners of a focal country
Node	Element of the network, in this case a country
Normalized brokerage	In ego analysis, brokerage divided by number of pairs
Number of directed ties	The number of connections among all the nodes in the ego network
Number of ordered pairs	The number of possible directed ties in each ego network
Outdegree	Measure of outgoing ties (number of ties or cumulative strength)
outdegree - activity	SIENA effect. Positive parameter estimate reflect propensity of nodes with high outdegree to form ties to others. Has equivalent meaning for indegree-activity effect
Quadratic Assignment Procedure (QAP)	Equivalent to regression analysis for non-network data See Hubert and Schultz (1976) ; Krackhardt (1987)
Reciprocity	Share of reciprocated ties. As SIENA effect, reflect tendency towards reciprocated ties
Regular equivalence	Regular equivalence may be understood as a partition of nodes into classes such that nodes of the same class are surrounded by the same classes of nodes. See White and Reitz (1983) ; Borgatti and Everett (1989)
Size of ego network	The number of nodes that one-step out neighbors of ego, plus ego itself
Structural equivalence	Two objects are structurally equivalent if they have the same ties to the same objects. See Lorrain and White (1971)

Appendix C. Multi-dimensional scaling of QAP correlations of annual adjacency matrices for the truffle trade in the 1988–2006 period (arrows are superimposed).



Appendix D. Grouped Adjacency Matrix of Factions routine for 2006 trade network of truffles, US\$ 1000.

	Malaysia	Morocco	Canada	Indonesia	USA	Sweden	Italy	France	United Kingdom	Belgium
Malaysia				26.699						
Morocco			57.944		48.010			474.193	1.328	
Canada										
Indonesia					29.850	14.000	20.800			
USA			142.422	1.351			3.143			
Sweden					0.815					
Italy	3.330		158.406		1811.453	37.362		2813.147	516.385	504.008
France	11.298		85.191		1505.472	73.365	198.344	20.088	199.052	573.691
United Kingdom						33.829				
Belgium						13.722		10.727	149.036	
Spain					22.652			1546.366	39.633	0.247
Switzerland							2.681			
Netherlands					3.024	0.136		7.533	0.029	6.780
Germany			10.066				6.416	2.511		5.911
China, Hong Kong SAR										
China	16.272		15.184		21.725			26.367	63.002	3.576
Singapore										
Australia										
Rep. of Korea										
Japan										
	Spain	Switzerland	Netherlands	Germany	China, Hong Kong SAR	China	Singapore	Australia	Rep. of Korea	Japan
Malaysia							0.537			
Morocco	53.842									
Canada										
Indonesia			10.546	10.912						
USA										
Sweden										
Italy	570.534	970.280	962.325	1359.723	66.637	5.124	70.718	126.708	15.118	99.834
France	259.448	165.436	109.215	720.565	47.427		48.958	22.177	38.916	917.654

United Kingdom			70.353						
Belgium	1.680		36.246	10.143					
Spain				4.961					37.816
Switzerland	2.265								
Netherlands				91.000					
Germany	0.847	7.217	22.613						
China, Hong Kong SAR									
China			5.250	409.480	85.808		6.048	34.995	856.549
Singapore									
Australia				0.282			3.805		
Rep. of Korea									
Japan									

Appendix E. Ego network analysis of truffle trade in 2006.

	Export								
	Dichotomized at $x > 0$			Dichotomized at $x > \text{US\$ } 4832$ (median)			Dichotomized at $x > \text{US\$ } 63,476$ (mean)		
	Size	Density	Normalized brokerage	Size	Density	Normalized brokerage	Size	Density	Normalized brokerage
France	53	0.05	0.95	37	0.04	0.96	12	0.10	0.90
Italy	67	0.04	0.96	41	0.05	0.95	17	0.05	0.95
China	26	0.10	0.90	21	0.08	0.92	3	0.00	1.00
Japan	1			0	0.00	0.00	0	0.00	0.00
Germany	18	0.18	0.82	7	0.21	0.79	0	0.00	0.00
Spain	18	0.06	0.94	11	0.06	0.94	1		
USA	24	0.02	0.98	7	0.02	0.98	1		
China, Hong Kong SAR	2	0.00	1.00	0	0.00	0.00	0	0.00	0.00
Switzerland	3	0.33	0.67	1			0	0.00	0.00
United Kingdom	10	0.03	0.97	3	0.00	1.00	1		
Netherlands	15	0.14	0.86	6	0.23	0.77	1		
Belgium	14	0.16	0.84	9	0.17	0.83	1		
Morocco	6	0.30	0.70	5	0.35	0.65	1		
Malaysia	4	0.08	0.92	1			0	0.00	0.00
Singapore	3	0.00	1.00	0	0.00	0.00	0	0.00	0.00
Canada	1			0	0.00	0.00	0	0.00	0.00
Sweden	2	0.00	1.00	0	0.00	0.00	0	0.00	0.00
Indonesia	7	0.36	0.64	7	0.24	0.76	0	0.00	0.00
Rep. of Korea	0	0.00	0.00	0	0.00	0.00	0	0.00	0.00
Australia	6	0.03	0.97	0	0.00	0.00	0	0.00	0.00
	Import								
	Size	Density	Normalized brokerage	Size	Density	Normalized brokerage	Size	Density	Normalized brokerage
	Size	Density	Normalized brokerage	Size	Density	Normalized brokerage	Size	Density	Normalized brokerage
France	8	0.38	0.63	6	0.27	0.73	3	0.17	0.83
Italy	6	0.33	0.67	4	0.25	0.75	1		
China	1	0.00		1	0.00		0	0.00	0.00
Japan	6	0.37	0.63	5	0.40	0.60	3	0.00	1.00
Germany	11	0.31	0.69	10	0.30	0.70	5	0.20	0.8
Spain	6	0.53	0.47	3	0.50	0.50	2	1.00	0.00
USA	12	0.20	0.80	10	0.18	0.82	3	0.33	0.67
China, Hong Kong SAR	6	0.23	0.77	5	0.25	0.75	3	0.00	1.00
Switzerland	6	0.47	0.53	4	0.42	0.58	3	0.33	0.67

United Kingdom	12	0.25	0.75	6	0.37	0.63	4	0.33	0.67
Netherlands	7	0.50	0.50	7	0.45	0.55	3	0.67	0.33
Belgium	8	0.46	0.54	5	0.65	0.35	2	1.00	0.00
Morocco	0	0.00	0.00	0	0.00	0.00	0	0.00	0.00
Malaysia	4	0.42	0.58	3	0.33	0.67	0	0.00	0.00
Singapore	6	0.37	0.63	4	0.42	0.58	1		
Canada	11	0.27	0.73	8	0.30	0.70	3	0.67	0.33
Sweden	8	0.45	0.55	6	0.40	0.60	1		
Indonesia	2	0.00	1.00	1	0.00		0	0.00	0.00
Rep. of Korea	3	0.67	0.33	3	0.67	0.33	0	0.00	0.00
Australia	3	0.33	0.67	2	1.00	0.00	1		

Appendix F. Analysis of structural holes for truffle trade in 2006.

	Degree	Effective size	Efficiency	Constraint	Hierarchy
Dichotomized at $x > \text{US\$ } 0$ (i.e. all existing trade flows are recoded to 1).					
France	55	50.008	0.909	0.088	0.288
Italy	68	63.178	0.929	0.067	0.250
China	27	22.204	0.822	0.151	0.150
Japan	7	4.214	0.602	0.469	0.040
Germany	22	16.552	0.752	0.190	0.166
Spain	21	18.188	0.866	0.169	0.240
USA	33	30.333	0.919	0.096	0.186
China, Hong Kong SAR	8	5.688	0.711	0.366	0.079
Switzerland	8	5.167	0.646	0.427	0.150
United Kingdom	20	16.659	0.833	0.186	0.179
Netherlands	18	13.091	0.727	0.221	0.155
Belgium	17	12.682	0.746	0.239	0.187
Morocco	6	3.833	0.639	0.536	0.112
Malaysia	8	4.688	0.586	0.409	0.047
Singapore	9	6.444	0.716	0.306	0.051
Canada	12	8.667	0.722	0.278	0.076
Sweden	10	6.600	0.660	0.367	0.068
Indonesia	8	5.444	0.681	0.428	0.132
Rep. of Korea	3	1.333	0.444	0.997	0.015
Australia	9	7.389	0.821	0.296	0.118
Dichotomized at $x > \text{US\$ } 4832$ (median)					
France	39	35.314	0.905	0.116	0.314
Italy	42	38.244	0.911	0.104	0.285
China	22	18.705	0.850	0.165	0.187
Japan	5	3.000	0.600	0.615	0.037
Germany	13	9.176	0.706	0.301	0.165
Spain	13	10.893	0.838	0.253	0.224
USA	16	13.853	0.866	0.183	0.130
China, Hong Kong SAR	5	3.600	0.720	0.507	0.063
Switzerland	5	3.700	0.740	0.545	0.129
United Kingdom	9	6.000	0.667	0.388	0.093
Netherlands	10	6.269	0.627	0.352	0.129
Belgium	11	7.893	0.718	0.328	0.171
Morocco	5	3.000	0.600	0.624	0.097
Malaysia	4	3.000	0.750	0.535	0.092
Singapore	4	2.250	0.563	0.730	0.035
Canada	8	4.813	0.602	0.422	0.062

Sweden	6	3.333	0.556	0.548	0.042
Indonesia	8	6.375	0.797	0.378	0.176
Rep. of Korea	3	1.333	0.444	0.997	0.015
Australia	2	1.000	0.500	1.389	0.000
Dichotomized at $x > \text{US\$ } 63,476$ (mean)					
France	13	11.000	0.846	0.248	0.309
Italy	17	15.278	0.899	0.174	0.317
China	3	3.000	1.000	0.333	0.000
Japan	4	3.500	0.875	0.472	0.083
Germany	5	4.000	0.800	0.439	0.076
Spain	2	1.000	0.500	1.235	0.057
USA	4	2.750	0.688	0.623	0.063
China, Hong Kong SAR	3	3.000	1.000	0.333	0.000
Switzerland	3	2.333	0.778	0.728	0.077
United Kingdom	5	3.400	0.680	0.538	0.079
Netherlands	4	2.000	0.500	0.779	0.051
Belgium	3	1.333	0.444	0.997	0.015
Morocco	1	1.000	1.000	1.000	1.000
Malaysia	0	0.000			
Singapore	1	1.000	1.000	1.000	1.000
Canada	3	1.333	0.444	0.997	0.015
Sweden	1	1.000	1.000	1.000	1.000
Indonesia	0	0.000			
Rep. of Korea	0	0.000			
Australia	1	1.000	1.000	1.000	1.000

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